

ZIQI ZHANG

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EDUCATION

Georgia Institute of Technology, GA, USA <i>Ph.D. in Computational Science and Engineering</i> • Advisor: Prof. Xiuwei Zhang	<u>Sep 2020 - Dec 2025</u>
Georgia Institute of Technology, GA, USA <i>Master of Computational Science and Engineering</i> • Advisor: Prof. Xiuwei Zhang	<u>Sep 2020 - Dec 2023</u>
Georgia Institute of Technology, GA, USA <i>Graduate Study in Electrical and Computer Engineering</i> <i>Transitioned to Ph.D. program in Computational Science and Engineering</i>	<u>Sep 2019 - Aug 2020</u>
Beihang University, Beijing, China <i>Bachelor of Electronic and Information Engineering</i>	<u>Sep 2015 - Jul 2019</u>

WORK EXPERIENCE

Assistant Professor [intended] <i>Cheriton School of Computer Science, University of Waterloo, ON, Canada</i>	<u>Sep 2027 - Aug 2029</u>
Affiliated Faculty [intended] <i>Vector Institute, ON, Canada</i>	<u>Sep 2027 - Aug 2029</u>
Postdoctoral Fellow <i>Georgia Institute of Technology, GA, USA</i>	<u>Jan 2026 - Aug 2027</u>

PUBLICATION

Journal & Conference papers

1. **Ziqi Zhang**, Jongseok Han, Le Song, and Xiuwei Zhang[†]. “CeSpGRN: Inferring cell-specific gene regulatory networks from single cell multi-omics and spatial data”, *Bioinformatics*, 2026
2. Yuqi Cheng, Boshuai Zhang, Chandra Sekhar Reddy Edula, Mo Sun, **Ziqi Zhang**, Xiuwei Zhang[†]. “Mixture-of-Residuals Meets LLM-Semantic Gene Modulation: Spatial Transcriptomic Inference from H&E Images”, *KDD*, accepted as oral presentation, 2026
3. Hechen Li, **Ziqi Zhang**, Michael Squires, Xi Chen, Xiuwei Zhang[†], “scMultiSim: simulation of multi-modality single cell data guided by cell-cell interactions and gene regulatory networks”, *Nature Methods*, 2025
4. **Ziqi Zhang**, and Xiuwei Zhang[†]. “Data-driven batch detection enhances single-cell omics data analysis.” *Cell Systems* 2024.
5. **Ziqi Zhang**, Xinye Zhao, Mehak Bindra, Peng Qiu, and Xiuwei Zhang[†]. “scDisInFact: disentangled learning for integration and prediction of multi-batch multi-condition single-cell RNA-sequencing data”, *Nature Communications*, 2024
6. **Ziqi Zhang**, Haoran Sun, Ragunathan Mariappan, Xinyu Chen, Xi Chen, Mika S Jain, Mirjana Efremova, Sarah A Teichmann, Vaibhav Rajan, Xiuwei Zhang[†]. “scMoMaT jointly performs single cell mosaic integration and multi-modal bio-marker detection”, *Nature Communications*, 2023
7. **Ziqi Zhang**, Chengkai Yang, and Xiuwei Zhang[†]. “scDART: integrating unmatched scRNA-seq and scATAC-seq data and learning cross-modality relationship simultaneously” *Genome Biology*, 2022

8. **Ziqi Zhang** and Xiuwei Zhang[†]. “Inference of high-resolution trajectories in single-cell RNA-seq data by using RNA velocity”, *Cell Reports Methods*, 2021
9. Vaibhav Rajan, **Ziqi Zhang**, Carl Kingsford, and Xiuwei Zhang[†]. “Maximum likelihood reconstruction of ancestral networks by integer linear programming”, *Bioinformatics*, 2021

Preprints & Working papers

1. Macrina Lobo, **Ziqi Zhang**[†], and Xiuwei Zhang[†]. “Spatiotemporal cell type deconvolution leveraging tissue structure”, *Major Revision at Nature Communications*, 2026
2. **Ziqi Zhang**, Chandra Sekhar Reddy E, Chao Zhang, and Xiuwei Zhang[†]. “scREBOUND: An Efficient Design of single-cell Foundation Model with Batch Representation”, *BioRxiv*, 2026
3. Hongshun Shi, **Ziqi Zhang**¹, Lourdes Brea, Zhuoyuan Lin, Viriya Keo, Sambhavi Senthil, BaoHan Thi Vo, Galina Gritsina, Haydn Thomas Kissick, Jonathan Zhao, Xiuwei Zhang, Jindan Yu[†]. “CXCR7 mediates an immune-suppressive tumor microenvironment and promotes prostate cancer progression”, *working paper*, 2026
4. **Ziqi Zhang** and Xiuwei Zhang[†]. “Velosim: Simulating single cell gene-expression and RNA velocity”, *BioRxiv*, 2021

Workshop

1. **Ziqi Zhang** and Xiuwei Zhang. “Integrating unmatched scRNA-seq and scATAC-seq data and learning cross-modality relationship simultaneously” **Spotlight**, *16th Machine Learning in Computational Biology (MLCB)*, 2021
2. **Ziqi Zhang**, Haoran Sun, Ragunathan Mariappan, Xi Chen, Mika S Jain, Mirjana Efremova, Sarah A Teichmann, Vaibhav Rajan, Xiuwei Zhang. “Integrating unpaired scRNA-seq and scATAC-seq with unequal cell type compositions” *ICML workshop on Computational Biology*, 2021

Talks

1. Talk: *Scalable Representation Learning for Multi-View, Multi-Condition, Single-Cell Data*, the University of Waterloo, Canada, 2026
2. Talk: *Scalable Representation Learning for Multi-View, Multi-Condition, Single-Cell Data*, City University of Hong Kong, Hong Kong, 2026
3. Poster presentation: *scDisInFact: the disentangled learning for integration and prediction of multi-batch multi-condition single-cell RNA sequencing data*, MCBIOS, 2024 [**Best poster presentation award**]
4. Poster presentation: *CeSpGRN: Inferring cell-specific gene regulatory network from single cell multi-omics data and spatial data*, CSHL Single Cell Analysis, 2023
5. Talk: *Inferring GRNs and CCIs using single-cell sequencing data*, InQuBETA workshop: Computational Problems in Single Cell Transcriptomics and Multi-omics, 2023
6. Talk: *scDisInFact: the disentangled learning for integration and prediction of multi-batch multi-condition single-cell RNA sequencing data*, AWSOM-Atlanta Workshop on Single-cell Omics, 2023 [**Best oral presentation award**]
7. Poster presentation: *scDisInFact: the disentangled learning for integration and prediction of multi-batch multi-condition single-cell RNA sequencing data*, RSGDREAM, 2022
8. Talk: *scDisInFact: the disentangled learning for integration and prediction of multi-batch multi-condition single-cell RNA sequencing data*, HotCSE, School of Computational Science and Engineering, Georgia Institute of Technology, 2022
9. Talk: *scMoMaT: Multimodal Data Integration Software*, AWSOM-Atlanta Workshop on Single-cell Omics, 2022
10. Talk: *Integrating unmatched scRNA-seq and scATAC-seq data and learning cross-modality relationship simultaneously*, MLCB, 2021 [**Spotlight presentation**]

[†]Underline means equal contribution

11. Flash talk: *Learning latent embedding of multi-modal single cell data and cross-modality relationship simultaneously*, CSHL, 2021
12. Poster presentation: *CeSpGRN: Inferring cell-specific gene regulatory networks from single cell gene expression data*, RECOMB, 2022
13. Poster presentation: *scMoMaT: Integrating arbitrary number of single-cell multi-omics matrices using matrix-trifactorization*, USC-QCB, 2022
14. Poster presentation: *Inference of multiple trajectories in single-cell RNA-Seq data from RNA-velocity*, VIB, 2020
15. Poster presentation: *Inference of multiple trajectories in single-cell RNA-Seq data from RNA-velocity*, RSGDREAM, 2020
16. Poster presentation: *Inference of multiple trajectories in single-cell RNA-Seq data from RNA-velocity*, CSHL-Asia, 2020
17. Flash talk: *Computational methods to study cell mechanisms from single cell data*, Georgia Scientific Computing Symposium (GSCS), 2020

TEACHING

Courses

1. **Guest Lecture:** *CSE 6140 Algorithm* 2022Fall
2. **Head Teaching Assistant:** *CSE 6140 Algorithm* 2021Fall
3. **Teaching Assistant:** *CSE 8803 Machine learning in Computational Biology* 2021Spring
4. **Teaching Assistant:** *CSE 6140 Algorithm* 2020Fall

Advisee

1. **Macrina M Lobo:** Ph.D. student Sep 2024-Now
Project: *SpaDecoder: Decoding Spatial Transcriptomes at Single Cell Resolution Leveraging Tissue Structure*
2. **Chandra Sekhar Reddy E:** Master student Sep 2023-Now
Project: *Efficient design of single-cell foundation model*
3. **Xinye Zhao:** Master student Nov 2021-Oct 2022
Project: *scDisInFact: disentangled learning for integration and prediction of multi-batch multi-condition single cell RNA-sequencing data*
4. **Xinyu Chen:** Ph.D. student Aug 2021-Nov 2021
Project: *scMoMaT: Mosaic integration of single cell multi-omics data using matrix tri-factorization*
5. **Jongseok Han:** Master student Jun 2021-Nov 2021
Project: *Inferring cell-specific gene regulatory networks from single cell gene expression data*
6. **Chengkai Yang:** undergraduate student Jun 2020-Jan 2021
Project: *Learning latent embedding of multi-modal single cell data and cross-modality relationship simultaneously*

SERVICE

Program Committee:	<i>ISMB2026, MLCB 2025</i>
Journal reviewer:	<i>Nature Communications, Patterns, Bioinformatics, Cell Reports Methods, Bioinformatics Advances, BMC Bioinformatics, BMC Endocrine Disorders, Journal of Agricultural, Biological, and Environmental Statistics</i>
Conference reviewer:	<i>RECOMB2024, MLCB2023, ISMB2023, RECOMB 2022, ISMB 2022</i>